



CURRICULUM OVERVIEW

AI Builder Path

Build 3 Real AI Projects

A complete, project-driven journey from first line of Python to real machine learning, computer vision, and large language model applications.

Students don't just learn about AI — they write the code, train the models, and ship three working portfolio projects.

3

Courses

24

Lessons Total

3

Portfolio Projects

12–18

Age Range

LEARNING PROGRESSION

04

Python & AI Foundations

Real Python from scratch, then a first machine learning model.

05

Computer Vision Projects

Apps that see: face detection, emotion recognition, object detection.

06

Language Models & Chatbots

Real LLM apps with memory, personality, and a live user interface.

Learn by building, not by watching. Every lesson pairs a real concept with real, runnable code — no shallow demos, no black boxes.

A portfolio, not just a certificate. Students finish with three working AI applications they can show, explain, and keep building on.

COURSE 04 · LESSONS 1–8

Python and AI Foundations

Write Python scripts and understand how machine learning really works

Python 3

Jupyter Notebooks

scikit-learn

NumPy

OVERVIEW

The foundation of everything. Your child learns Python from scratch — variables, loops, functions, data — then builds their first machine learning model. No fluff, just real code.

PORTFOLIO PROJECTS

- Number prediction model
- Simple image classifier
- Data pattern finder

WHY IT MATTERS

Every machine learning system — from spam filters to recommendation engines — follows this same workflow: prepare data, train a model, test it honestly, and explain the result. This course is the foundation every later project builds on.

LESSON BREAKDOWN

- 1 Python setup and first program
- 2 Variables, data types, and logic
- 3 Loops, functions, and modules
- 4 Working with data and lists
- 5 Introduction to machine learning concepts
- 6 Building a first prediction model
- 7 Training, testing, and evaluating models
- 8 Presenting results to non-technical audiences

COURSE 05 · LESSONS 1–8

Computer Vision Projects

Build apps that see, detect, and recognize the real world

Python

OpenCV

TensorFlow / Keras

Webcam API

OVERVIEW

Your child builds apps that use a camera to detect faces, recognize objects, and read emotions — the same technology used in self-driving cars and medical diagnostics.

PORTFOLIO PROJECTS

- Real-time emotion detector
- Object recognition app
- Motion-activated security cam

WHY IT MATTERS

Computer vision powers everyday technology students already use — unlocking a phone with a face, sorting photos, or a car spotting a pedestrian. Students learn how these systems actually work, including where they can fail.

LESSON BREAKDOWN

- 1 How computers see images (pixels and arrays)
- 2 OpenCV setup and first image processing
- 3 Face detection with pre-trained models
- 4 Emotion recognition and classification
- 5 Object detection with YOLO
- 6 Real-time webcam processing
- 7 Building a full computer vision app
- 8 Deploying and demoing your project

COURSE 06 · LESSONS 1–8

Language Models and Chatbots

Build conversational AI apps using real LLM APIs

Python

OpenAI API

LangChain

Streamlit

OVERVIEW

Your child learns how ChatGPT actually works, then builds their own chatbot applications using the OpenAI API — complete with memory, personality, and real-world purpose.

PORTFOLIO PROJECTS

- Recipe chatbot with memory
- Study assistant bot
- Creative story generator

WHY IT MATTERS

Large language models now power search, customer support, and everyday writing tools. Students learn what's actually happening behind the chat window, then build and ship their own AI-powered application with a real interface.

LESSON BREAKDOWN

- 1 How language models work (transformers simplified)
- 2 OpenAI API setup and first call
- 3 Prompt engineering fundamentals
- 4 Building conversational memory
- 5 Giving your bot a personality and purpose
- 6 Connecting to external data sources
- 7 Building a UI with Streamlit
- 8 Deployment and sharing

The Complete Journey

Across 3 courses and 24 lessons, students progress from writing their first Python program to building and deploying full AI applications — walking away with three real, working projects for their portfolio.

COURSE 04

Python & AI Foundations

- Core Python: variables, logic, loops, functions
- Working with real data and lists
- Core ML concepts: training, testing, evaluation
- Presenting results clearly

COURSE 05

Computer Vision

- How images become data (pixels & arrays)
- Face detection & emotion recognition
- Object detection with YOLO
- Real-time webcam applications

COURSE 06

Language Models & Chatbots

- How LLMs like ChatGPT work
- Prompt engineering & conversational memory
- Giving an AI personality and purpose
- Shipping a real UI with Streamlit

WHAT STUDENTS BUILD

1

Foundations Portfolio

Number prediction model · Simple image classifier · Data pattern finder

2

Computer Vision Portfolio

Real-time emotion detector · Object recognition app · Motion-activated security cam

3

LLM & Chatbot Portfolio

Recipe chatbot with memory · Study assistant bot · Creative story generator

BY THE END OF THE PATH, STUDENTS CAN



Write real Python programs and explain how a machine learning model is trained, tested, and evaluated.



Build a computer vision application that detects faces and objects in real time from a webcam.



Build and deploy a conversational AI application powered by a real large language model API.

Ready to start building?

All lessons and projects are taught live by an expert Lumexa instructor.